

How to train your model: Paradigms for training LLMs and how to know which one to use

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5. In-context Learning (ICL)
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Primer on Transformers:

Tokenisation

The process of segmenting text into "tokens" (not necessarily words)

Data-driven method that identifies **subwords**:

- can be words (e.g., *"into"*, *"cat"*)

- can be meaning-bearing units like morphemes (e.g., *"er"*, *"est"*)

- can be arbitrary substrings (e.g., *"ene"*, *"xyl"*)

Why tokenise?

- Tokens form a model's vocabulary (with a fixed size, e.g., 100k+)

- Even rare words can be represented (e.g., *"Eataly"* can be represented as *"eat"*, *"aly"*)

Primer on Transformers:

Language modelling as next token prediction

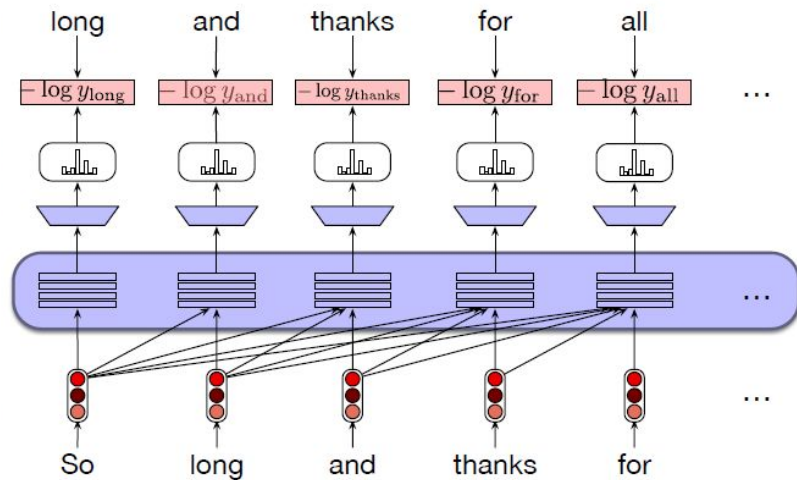
A transformer model is trained to predict the next token given the preceding text

Example:

"So long and thanks for ____"

The model assigns a **probability to each possible next token** (in the vocabulary)

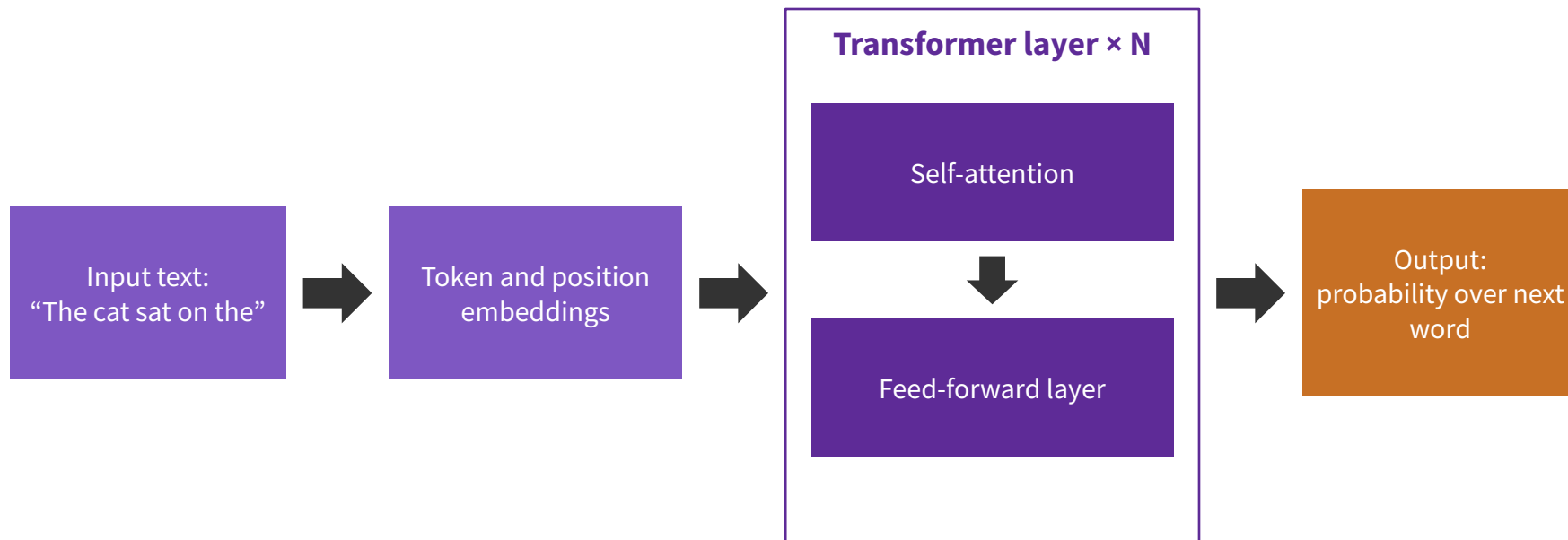
Trained on large amounts of text, the model acquires grammar, factual knowledge and stylistic patterns



Source: Jurafsky and Martin, *Speech and Language Processing*, 3rd edition

Primer on Transformers:

Inside a transformer



Primer on Transformers:

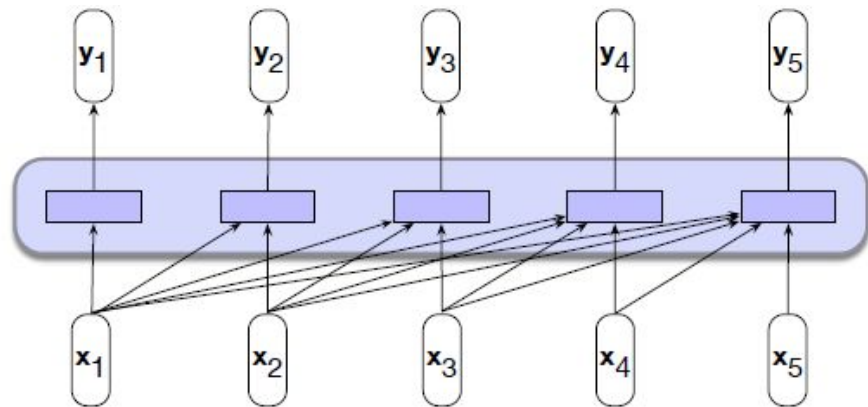
The role of self-attention

Each token forms a representation by **attending to every other word** in the context

In the sentence:

“The trophy did not fit in the suitcase because it was too big”

attention allows the model to resolve “it” as referring to the trophy rather than the suitcase



Source: Jurafsky and Martin, *Speech and Language Processing*, 3rd edition

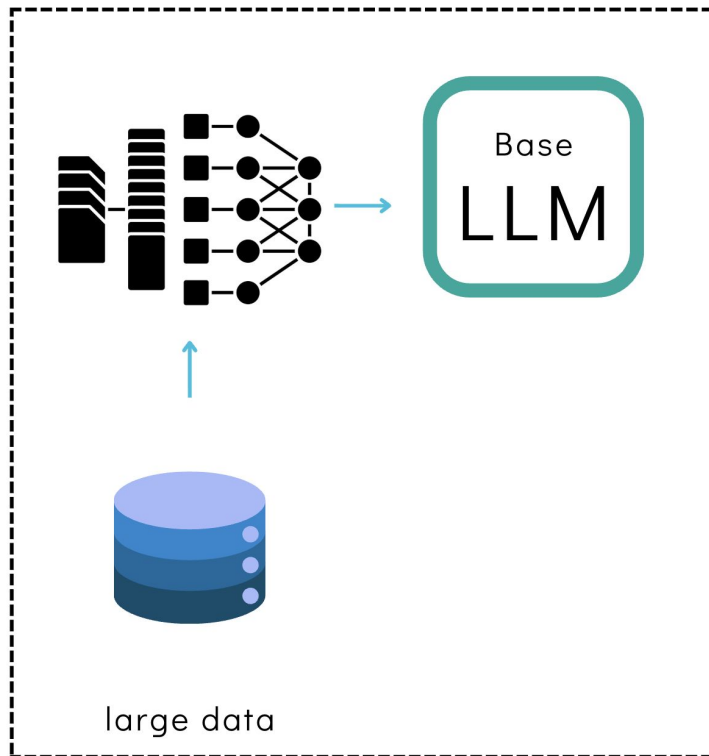
Model Pre-training

Training data consists of **large, broad text collections**

The training is "**self-supervised**": predicting the next token does not require any human-labelled examples

Pre-training is **very expensive**

The result is a **base model**: it predicts text well BUT it cannot follow any instructions nor perform any downstream task



Model Pre-training

Examples/Required Data

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Original paper

The association of mechanical versus manual chest compressions with neurological outcomes at hospital discharge among out-of-hospital cardiac arrests treated with intra-arrest transport

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Abstract

Aim: Mechanical chest compression (MCC) devices provide an opportunity to deliver high quality chest compressions during times when manual chest compressions may be challenging. Among non-traumatic out-of-hospital cardiac arrests (OHCA) treated with intra-arrest transport (IAT), we investigated the association of MCC, versus manual chest compressions, and patient outcomes.

Methods: Utilizing the British Columbia Cardiac Arrest Registry, we included adult OHCA cases (January 2019 – June 2025) who were transported to hospital with ongoing chest compressions. We classified patients as MCC or manual chest compressions. We fit an adjusted logistic regression model to estimate the association of MCC (versus manual chest compressions), and favorable neurological outcome at hospital discharge (Cerebral Performance Category 1–2).

Results: During the study period, there were 1967 cases treated with IAT, of whom 243 (12%) and 1724 (84%) received MCC and manual chest compressions, respectively. Patient and cardiac arrest characteristics between groups were similar, including presence of bystander CPR and any period of on-scene return of spontaneous circulation (ROSC). Comparing MCC and manual chest compression groups: 43 (18%) and 412 (24%) had ROSC at hospital arrival, 17 (7.0%) and 135 (7.8%) survived to hospital discharge, and 16 (6.6%) and 123 (7.1%) had favorable neurological outcomes, respectively. Our logistic regression model did not detect an association of MCC (versus manual chest compressions) with favorable neurological outcome at hospital discharge with (AOR 0.93, 95% CI 0.51–1.69).

Conclusion: Among cases treated with IAT, we did not detect an association between MCC, compared to manual chest compressions, and favorable neurological outcome at hospital discharge.

Keywords: Cardiac arrest, Cardiopulmonary resuscitation, Intra-arrest transport, Mechanical CPR, Mechanical chest compression, Out-of-hospital cardiac arrest

Introduction

Out-of-hospital cardiac arrest (OHCA) is common, with an incidence of approximately 80 per 100,000 residents.¹ Among OHCA patients that receive cardiopulmonary resuscitation (CPR) by emergency medical services (EMS), 8.8% survive to hospital discharge.² Previous studies have shown that high-quality CPR is a key determinant of patient survival.^{3–6}

The goal of CPR is to maintain perfusion and oxygenation throughout the body; interruptions in chest compressions lead to



Mechanical chest compression (MCC) devices provide an opportunity to deliver high quality chest compressions during times when manual chest compressions may be challenging.

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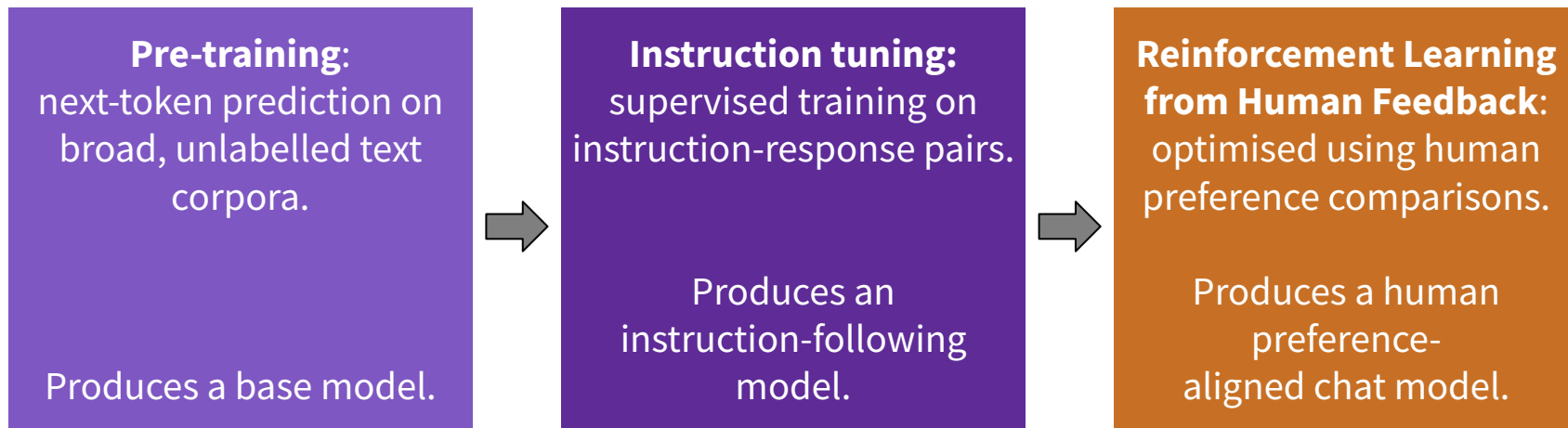
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Post-training:

How do we go from a base model to a chatbot-like model?



Post-training: Instruction Tuning

Base model

Prompt: *"Summarise this article in one sentence."*

Response: the model may continue the pattern, for example by listing further instructions rather than answering

The base model has only been trained to predict the next token, and not to treat the input as a request

Instruction-tuned model

Prompt: "Summarise this article in one sentence."

Response: a short summary

The model has been trained on a dataset containing paired instructions (requests) and responses

Instruction Tuning:

Examples/Required Data

Input (Instruction Given to Model)	Expected Model Output (Target Response)
<i>Explain how mechanical chest compression devices work</i>	<i>Mechanical chest compression devices compress the chest rhythmically to circulate blood when the heart has stopped. During cardiac arrest, the heart isn't pumping effectively, so compressions: push blood out of the heart to vital organs (brain, heart), and allow the chest to recoil so the heart refills. Mechanical devices replicate this cycle with precision.</i>
<i>Summarise the following doctor's note for me: [clinical note content]</i>	<i>Reason for Visit: Patient presented with a 3-day history of fever, sore throat, fatigue, and mild cough. Clinical Findings: Temperature: 38.2°C; Throat: Erythematous with mild swelling; Lungs: Clear on auscultation; No signs of respiratory distress Assessment: Likely viral upper respiratory tract infection.</i>

Supervised fine-tuning (SFT):

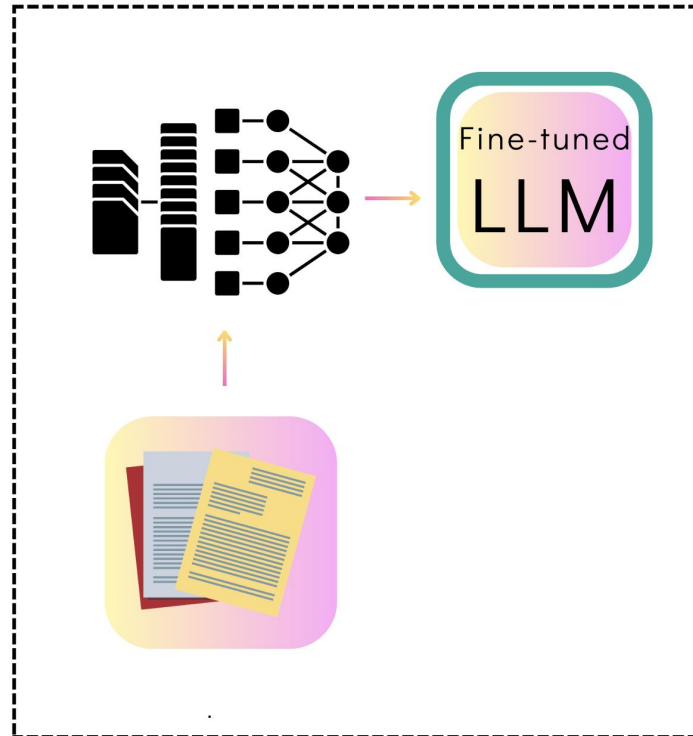
Making a model learn downstream tasks

The **aim**: adjust model behaviour/update model parameters without retraining from scratch

Requires **substantially less data** than pre-training: thousands to millions of examples rather than trillions of tokens.

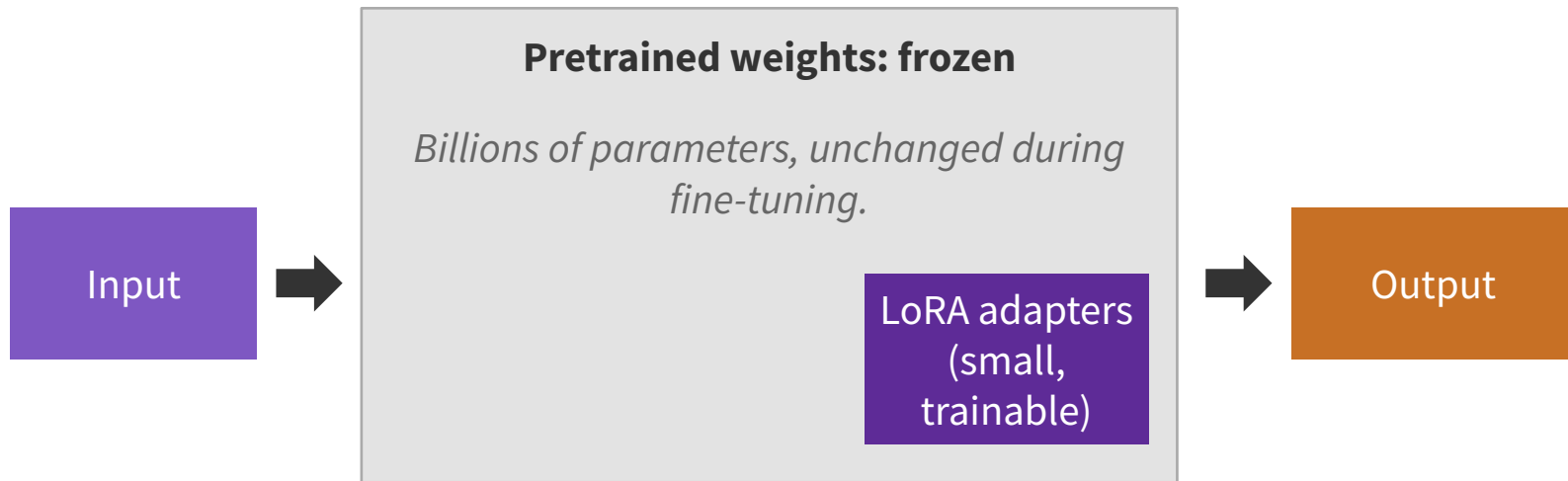
Example downstream tasks

- Text summarisation
- Machine translation
- Domain-specific question answering
- Text classification (e.g., sentiment analysis)
- Named entity recognition



Supervised fine-tuning (SFT):

Parameter-efficient fine-tuning (PEFT)



Full fine-tuning updates every parameter, requiring a full copy of the model per task.

PEFT methods such as LoRA (Low-Rank Adaptation) freeze the base model and train a small number of additional parameters.

Supervised fine-tuning (SFT):

Limitations of SFT

Relies on a set of reference answers (target response), but in many downstream tasks, there is no single correct response

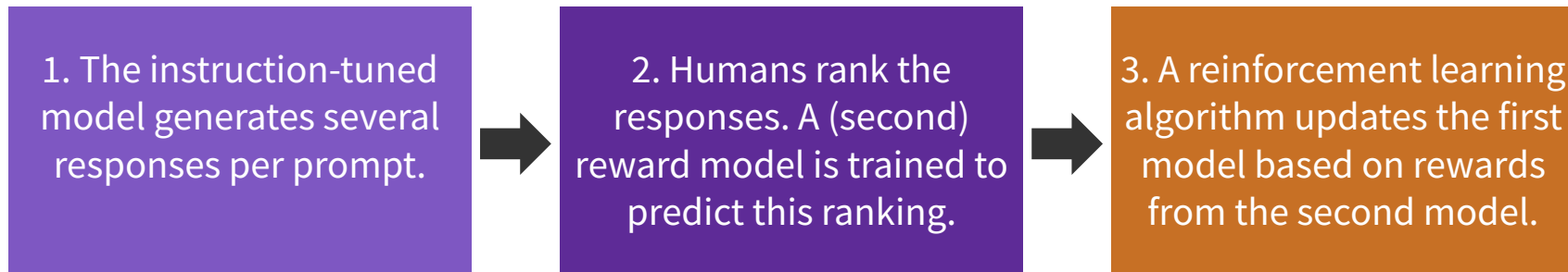
Identifying a poor response is often easier than writing an optimal one

An alternative is to learn from **comparisons between responses**, which are easier for humans to judge than to author

This comparison-based signal underpins **Reinforcement Learning from Human Feedback (RLHF)**

Post-training:

Reinforcement Learning from Human Feedback



This process can be repeated to refine the model iteratively.

Many current systems use simpler methods, such as **Direct Preference Optimization (DPO)**, that do not require a separate reward model.

RLHF:

Example/Required Data

Input	Response A	Response B	Preferred Response
<i>Explain how mechanical chest compression devices work</i>	<i>Mechanical chest compression devices compress the chest rhythmically to circulate blood when the heart has stopped. During cardiac arrest, the heart isn't pumping effectively, so compressions: push blood out of the heart to vital organs (brain, heart), and allow the chest to recoil so the heart refills. Mechanical devices replicate this cycle with precision.</i>	<i>Think of mechanical chest compression devices as automated CPR systems that take over the physical job of pressing on the chest during cardiac arrest—doing it with steady timing and force that's hard for humans to maintain.</i>	<i>Response A</i>

In-context learning (ICL): Prompting

Examples are provided directly within a prompt, without updating any model weights

Zero-shot prompting: the model is given only an instruction (no examples)

Few-shot prompting: two to five examples are included

The model **infers the pattern** from context and completes it accordingly

Comparison:

ICL:

affects only the current context and has no lasting effect on the model;

suitable for short-lived or rapidly changing tasks

Fine-tuning:

updates the model's parameters permanently

generally more effective where consistent behaviour change is required

In-context Learning (ICL) :

Limitations of ICL

Bounded by the model's context window: the amount of text it can process in a single prompt

Examples must be included in every request, since no parameters are updated between calls

Longer prompts increase cost

Performance can be sensitive to the choice, number and order of examples provided

Limitations of trained language models

A model's knowledge (embedded in its parameters) reflects only its training data and **has a fixed cut-off date**

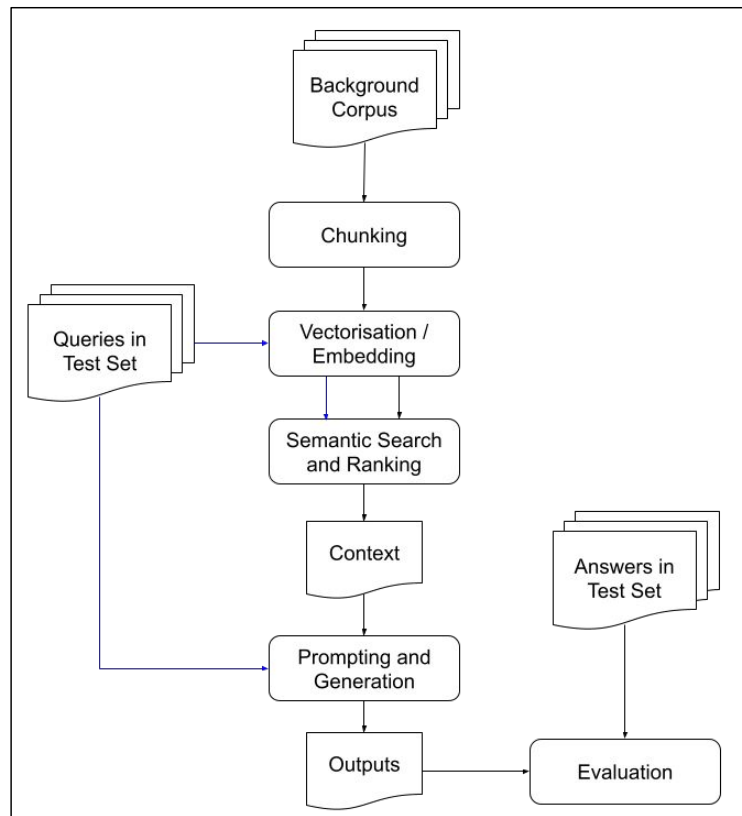
It has **no access to domain or organisation-specific information** unless this was included in training

When queried on unfamiliar topics, a model may produce plausible but incorrect information, referred to as **hallucination**

Retraining a model each time new information becomes available is impractical

Retrieval-Augmented Generation (RAG)

Supplies external context to the model without modifying its parameters



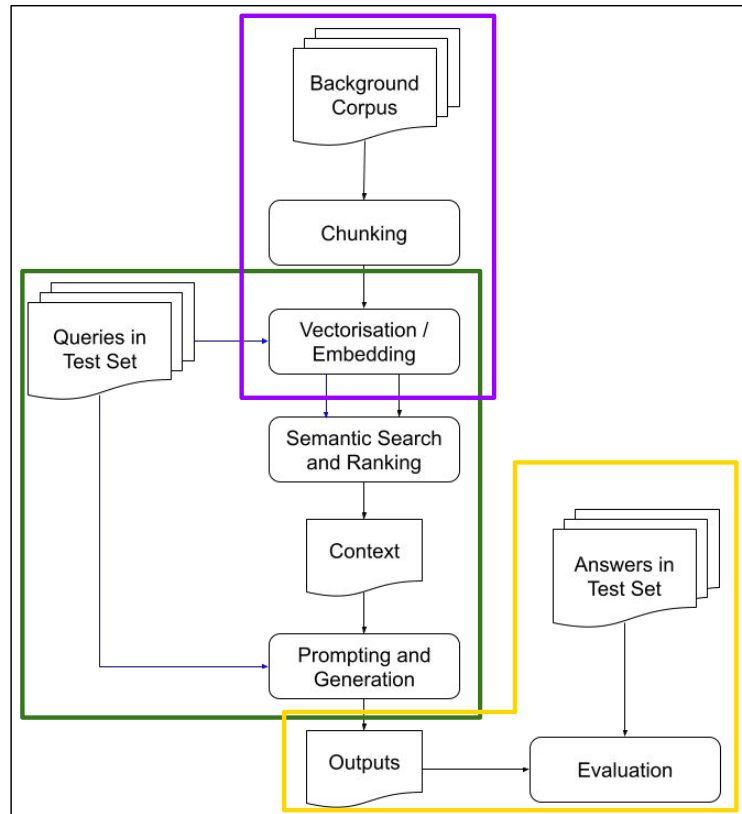
Retrieval-Augmented Generation (RAG)

Can be broken down into three stages:

Ingestion

Inference

Evaluation



Evaluating language models

Automatic benchmarks

Test sets that measure specific capabilities

Performance is reported as a score on held-out data the model was not trained on

Results are cheap to compute and easy to compare across models

Human-based evaluation

People read model outputs directly and judge them according to metrics (e.g., quality, helpfulness or safety).

Comparisons between two responses, as used in RLHF, are a form of human evaluation

More expensive than automatic benchmarks, but closer to how a model is actually used

Limitations of current evaluation practices

Popular benchmarks may not reflect the languages, domains or tasks a given application actually needs

Strong benchmark performance does not always translate into usefulness for a specific real-world task

Human evaluation is costly and can vary between annotators

A model can be optimised for a leaderboard without becoming more reliable in practice

Model choices:

Open-weight vs closed models

Open-weight models

Model weights are **published** and can be downloaded and run on infrastructure the user controls

Examples include **Llama**, **Mistral** and **Qwen**.

Full visibility of the architecture; the model can be inspected, modified, fine-tuned or run on a local machine

Closed models

Model weights are **not released**; the model is accessed only through an API or a hosted service

Examples include **ChatGPT**, **Claude** and **Gemini**.

Usage is typically simpler and includes managed infrastructure, but customisation is limited to what the provider exposes

Model choices:

Open-weight vs closed models

Open-weight models

Full fine-tuning and PEFT methods such as LoRA both require **direct access to model weights**

An open-weight model can be downloaded and **fine-tuned locally or on private infrastructure**

This is the better choice when a team needs **full control** over fine-tuning, data locality or deployment

Closed models

No weights are exposed, so **adaptation is limited** to what the provider allows

This typically means prompting, ICL or a hosted fine-tuning API restricted to the provider's own infrastructure and terms

LoRA and other PEFT methods are generally not available, since they require direct access to the model's parameters

Model choices:

Large vs small language models

Large language models

Typically tens to hundreds of billions of parameters

Broad general knowledge and strong performance across many tasks

Require substantial compute, typically served from cloud infrastructure

Higher cost and latency per query

Small language models

Typically one to ten billion parameters

Narrower capability, but lower cost, lower latency and able to run on local hardware

Suited to applications requiring on-device deployment or fast response times

Often fine-tuned for a single, well-defined task

Model choices:

When to choose a small language model (SLM)?

If any of the following applies to your use case:

- Strict **latency** requirements, for example real-time or voice interaction

- On-device or offline deployment, where **privacy or connectivity** is a constraint

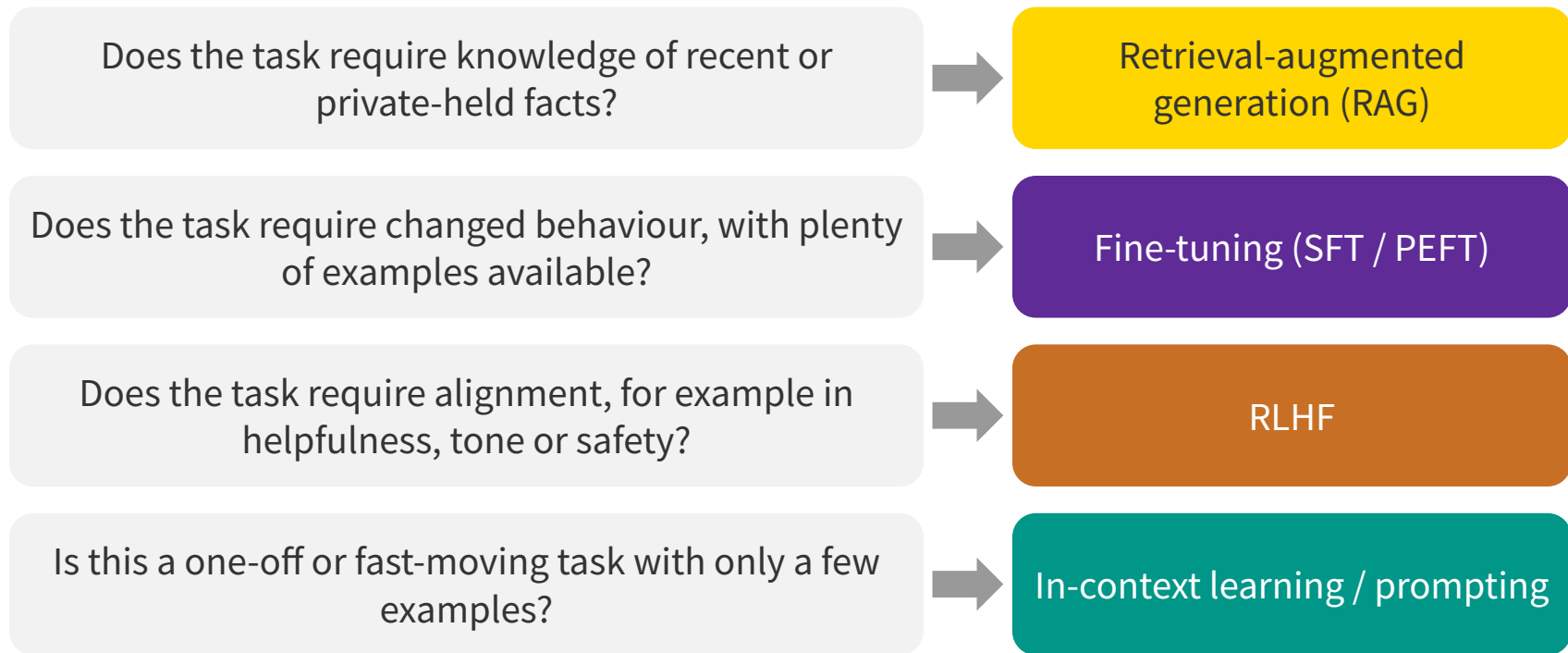
- High-query volume, where **inference cost** dominates overall cost

- A **narrow, well-defined** task that does not require broad general knowledge

Comparing the Different Training Paradigms

Paradigm	What changes	Data required	Relative cost	Best suited for
Pre-training	General language ability	Broad text corpora (100s of billions of tokens or more)	Very high	Foundation for all other paradigms
Instruction tuning	General instruction-following behaviour	Broad instruction-response datasets (thousands to millions)	Moderate	Turning a base model into a usable assistant
Supervised fine-tuning (SFT)	Behaviour specific to a downstream task	Labelled input-output pairs (thousands to millions)	Low to moderate	Specific downstream tasks
SFT with PEFT (e.g. LoRA)	Behaviour specific to a downstream task, with fewer trainable parameters	Same as supervised fine-tuning	Low	Fine-tuning with limited compute (open-weight models only)
RLHF	Alignment with human preferences	Human comparisons between responses	High	Chatbot-like assistants, alignment
ICL	Behaviour for a single prompt	A few examples in the prompt	Very low	One-off or fast-changing tasks
RAG	Access to external knowledge	A background corpus	Low to moderate	Up to date or private information

A simple framework for selecting a training paradigm



A simple framework for selecting a training paradigm

A support assistant over internal documentation that is updated weekly.

Retrieval-augmented generation, combined with light RLHF for tone and format

Classification of support tickets by urgency, with a large labelled dataset available.

Supervised fine-tuning of a small model

A one-off creative writing task, with no example data available and a short time frame.

In-context learning with a general-purpose model

An internal coding assistant that must run entirely on-premises

An open-weight model, fine-tuned locally with LoRA

Summary

Pre-training establishes a model's general language ability from large, unlabelled corpora.

Instruction tuning and **RLHF** turn a base model into an assistant that follows instructions and aligns with human preferences

Supervised fine-tuning makes a model learn a specific downstream task

In-context learning and **RAG** adapt model behaviour at inference time, without changing any parameters

These paradigms **are not mutually exclusive**: most deployed systems combine several of them

Whether a model's weights are open or closed determines whether fine-tuning methods such as LoRA are available at all, and the **choice of model and paradigm** overall depends on the task, the data available and the deployment constraints.

Thank you!

Find more about the work of the Artificial Intelligence and Text Mining for Social Good (ARTEMIS) Lab: <https://artemislabs.info/>

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